

# Agent-Based Modelling In Social Sciences

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## Final Exam

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Course director: Eduardo Tapia ([eduardo.tapia@liu.se](mailto:eduardo.tapia@liu.se))

Course Code: 771A21, 771A22

Student's name: \_\_\_\_\_

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## Instructions

- This exam must be completed individually. Copying is prohibited and will result in sanctions.
- Students are permitted to use any materials provided during the course.
- **Internet usage is not allowed** during the exam except for consulting the NetLogo Dictionary. If any student is found using the internet, they will be disqualified from the exam.
- The exam has a duration of 3 hours.
- In case of questions or doubts about the exam, students should ask the teacher.
- Each answer in the exam must be a unique NetLogo file, which must be saved in your personal computer.
- Once the exam is completed, students must zip the files together and send them via email to: [eduardo.tapia@liu.se](mailto:eduardo.tapia@liu.se), specifying their name in the subject line.
- The final results will be sent by email to each student within one week.
- There are 8 questions. Each correct answer will give the student 3 points. The maximum number of points is 24. There is no penalty for wrong answers.
- Double-check your code using prints as much as you can!

## Problem 1

Implementing a NetLogo model named 'problem\_1.nlogo' with the following characteristics:

One 'SETUP' button which must create:

- 15 wolves randomly distributed across the world.
- 15 sheep randomly distributed across the world.

One 'GO' button which must execute these procedures:

- Wolves and sheep move randomly, but the wolves move 10 steps at a time, and the sheep move one step at a time.
- If a wolf encounters a sheep on the same patch, the sheep disappears, and the wolf grows by 0.5 units.

Implementing a monitor to check the number of sheep in the model.  
The model stops when there are no sheep left in the system.

## Problem 2

Implementing a NetLogo model named 'problem\_2.nlogo' with the following characteristics:

One 'SETUP' button which must create:

- 50 green turtles in different patches, each with a list variable called 'friends'.

One 'GO' button which must execute these procedures:

- Turtles walk randomly.
- Turtles interact: if one turtle meets another turtle or turtles in her patch, the turtle selects one of them and adds their 'who' to her friend list.
- Turtles forget: Turtles can only remember the 'who' of the last 4 turtles they have met.